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GEOMETRY



Figure 10.1 *The School of Athens* by the Renaissance artist Raphael, painted between 1509 and 1511, depicts some of the greatest minds of ancient times. (credit: modification of work “School of Athens” by Raphael (1483–1520), Vatican Museums/Wikimedia, Public Domain)

Chapter Outline

- 10.1 Points, Lines, and Planes
- 10.2 Angles
- 10.3 Triangles
- 10.4 Polygons, Perimeter, and Circumference
- 10.5 Tessellations
- 10.6 Area
- 10.7 Volume and Surface Area
- 10.8 Right Triangle Trigonometry



Introduction

The painting *The School of Athens* presents great figures in history such as Plato, Aristotle, Socrates, Euclid, Archimedes, and Pythagoras. Other scientists are also represented in the painting.

To the ancient Greeks, the study of mathematics meant the study of geometry above all other subjects. The Greeks looked for the beauty in geometry and did not allow their geometrical constructions to be “polluted” by the use of anything as practical as a ruler. They permitted the use of only two tools—a compass for drawing circles and arcs, and an unmarked straightedge to draw line segments. They would mark off units as needed. However, they never could be sure of what the units meant. For instance, how long is an inch? These mathematicians defined many concepts.

The Greeks absorbed much from the Egyptians and the Babylonians (around 3000 BCE), including knowledge about congruence and similarity, area and volume, angles and triangles, and made it their task to introduce proofs for everything they learned. All of this historical wisdom culminated with Euclid in 300 BCE.

Euclid (325–265 BC) is known as the father of geometry, and his most famous work is the 13-volume collection known as *The Elements*, which are said to be “the most studied books apart from the Bible.” Euclid brought together everything offered by the Babylonians, the Egyptians, and the more refined contributions by the Greeks, and set out, successfully, to organize and prove these concepts as he methodically developed formal theorems.

This chapter begins with a discussion of the most basic geometric tools: the point, the line, and the plane. All other topics flow from there. Throughout the eight sections, we will talk about how to determine angle measurement and learn how to recognize properties of special angles, such as right angles and supplementary angles. We will look at the relationship of angles formed by a transversal, a line running through a set of parallel lines. We will explore the concepts of area and perimeter, surface area and volume, and transformational geometry as used in the patterns and rigid